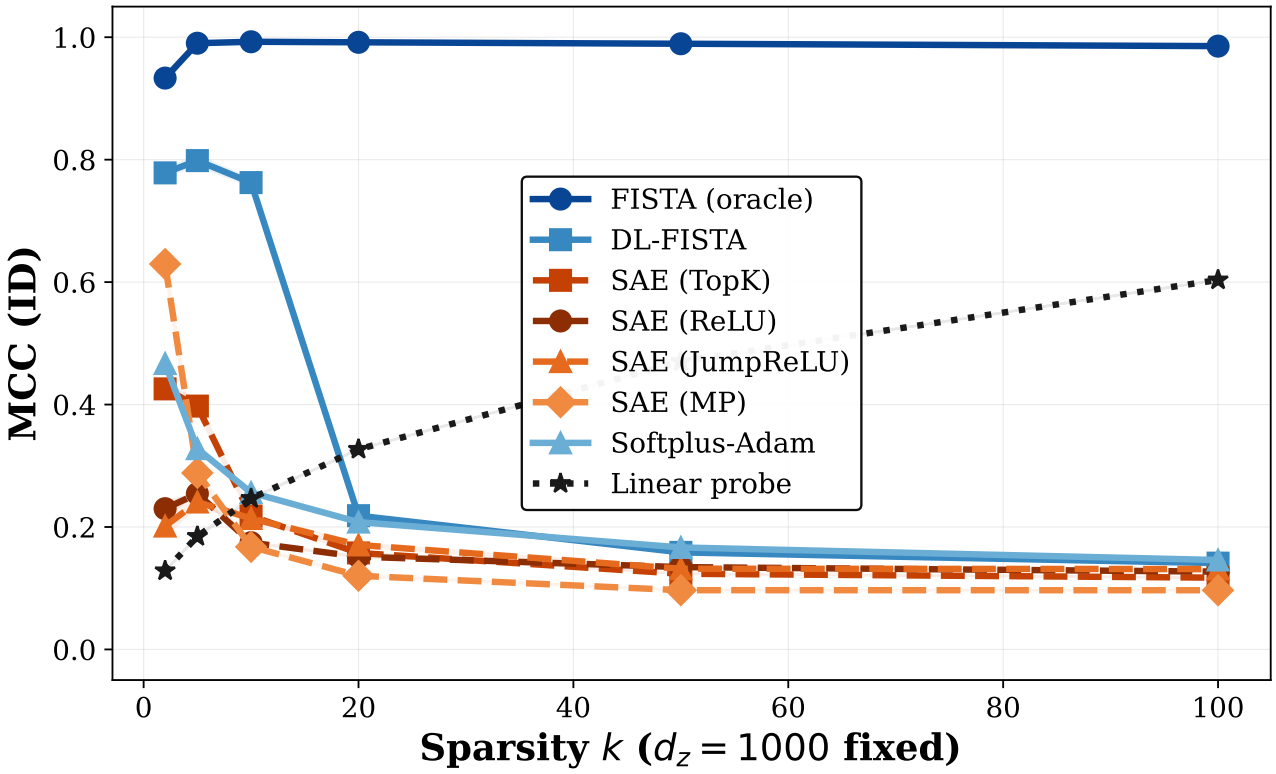
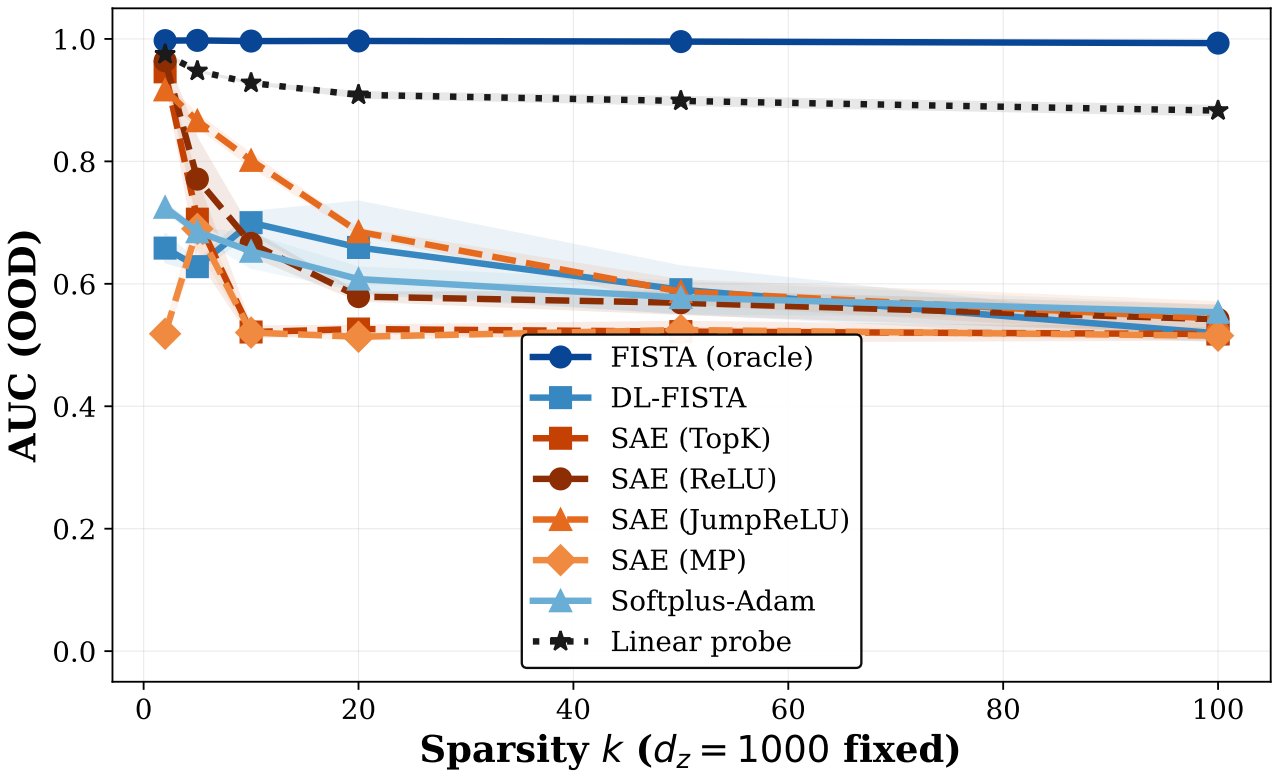


Per-sample methods degrade gracefully with sparsity; SAEs plateau regardless of k.

Figure 21: **Varying sparsity: MCC (ID).** Per-sample methods degrade gracefully with increasing k; SAEs plateau. $d_z = 1000$, d_y follows the CS bound.



OOD generalisation degrades with sparsity for SAEs; per-sample methods remain robust.

Figure 22: **Varying sparsity: AUC (OOD).** FISTA (oracle) remains near 1.0; all other methods degrade with increasing k, with SAEs and DL-FISTA converging toward 0.5. The linear probe degrades more gracefully (0.97 → 0.88) due to its supervised advantage. $d_z = 1000$.

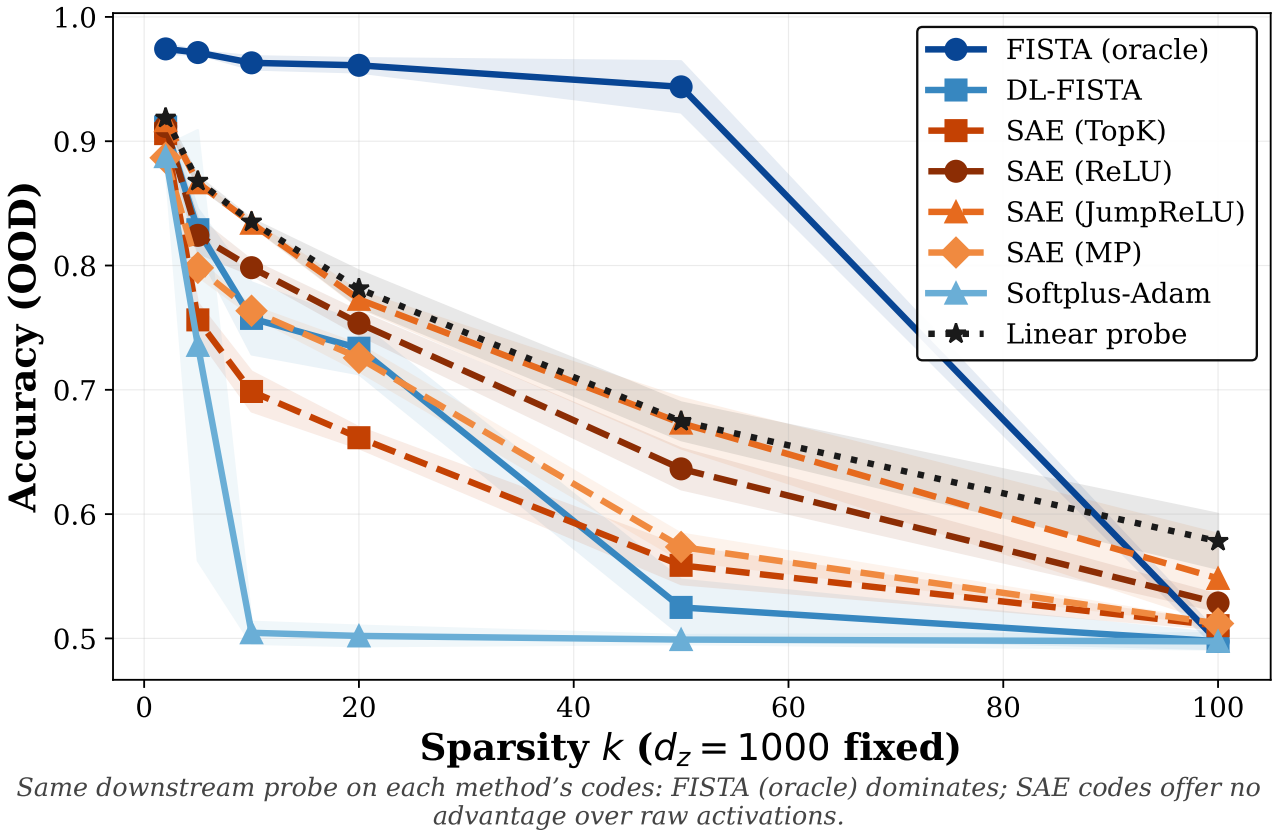


Figure 23: **Varying sparsity: Accuracy (OOD).** Same downstream probe on each method's codes. FISTA (oracle) dominates. DL-FISTA beats the linear probe at low k (≤ 10) but collapses at high k as dictionary learning fails. SAE codes offer no consistent advantage over raw activations. $d_z = 1000$.